Web of Science Practice

Zih-Yu Hsieh

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Question 1. *Provide the citation for a review article that cites Serlin, Tschirhart, Polshyn, Zhang, Zhu, Watanabe, Taniguchi, Balents and Young, "Intrinsic quantized anomalous Hall effect in a moiré heterostructure" Science 367:900-903 (2020). https://doi.org/10.1126/science.aay5533*

Proof. Here is one:

Li, Redekop, Beach, Zhang, Zhang, Liu, Holtzmann, Hu, Anderson, Park, Taniguchi, Watanabe, Chu, Fu, Cao, Xiao, Young and Xu, "Universal Magnetic Phases in Twisted Bilayer $MoTe_2$" Nano Letters 25:18044-18050 (2025). https://pubs.acs.org/doi/10.1021/acs.nanolett.5c04751 □

Question 2. *How many publications are referenced in the most cited paper co-authored by Vera Kistiakowsky?*

Proof. There are 23 publications referenced in the most cited paper co-authored by Vera Kistiakowsky. □

Question 3. *How many peer-reviewed articles cite Cornell & Wieman's first experimental report of trapping a Bose-Einstein condensate.*
Check that:

- *your answer counts only peer-reviewed articles.*
- *the report describes an experiment, not just its results (read the abstract, or the entire content).*
- *The report does not reference any earlier work on the same topic by the same authors.*

Proof. □

Question 4. *Find the oldest article in the American Journal of Physics measuring the speed of light.*

a. Proide its citation.

b. Describe its measurement approach.

Proof. The oldest article in AJP I found that's about measuring the speed of light, is "A laboratory experiment on the velocity of light" in 1954.

a. Citation:

   Palmer, Spratt, "A laboratory experiment on the velocity of light." Am. J. Phys. 22:481-485 (1954). https://doi.org/10.1119/1.1933791

b. The measurement of speed of light here requires a light source, and a Kerr cell shutter to produce light pulses. Then, they used a half-silvered mirror to divide the light pulses into two beams (in orthogonal directions). With the two beams traveling different distances, let them get reflected back, and combined to be measured by the deterctor. Then, by adjusting the distance each beam travels, knowing the frequency of the light pulses, and teh light's behavior detected (as sinusoidal waves), these data can be used to directly calculate the speed of light.

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